A Comparative Study of Statistical and Machine Learning Methods for Smart Data Imputation

B. Tech Project

Mid-evaluation Report

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Submitted by:
Dara Nissi Krithika
230002021

Guided by:
Prof. Trapti Jain

Co-supervised by:
Dr. Puneet Gupta



DEPARTMENT OF ELECTRICAL ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY INDORE

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Section 1

1.1 Introduction, Theme of work and Motivation

Smart meters are crucial for applications such as load forecasting, billing, demand-response analysis and distribution-system monitoring in smart-grid applications [1], [2]. Due to Sensor failures, communication problems, storage issues and other disruptions can result in missing observations in smart meters [1]–[3] and can consequently affect downstream analysis [4].

The ***theme*** of the project is therefore to perform a controlled comparative study of statistical, machine-learning and deep-learning approaches for recovering missing values in real smart-meter time-series data [1], [5], [6].

The ***motivation*** for the present project is the need for a systematic comparison of different classes of imputation techniques on real smart-meter electricity-consumption data [5]. A major challenge in the literature is that different imputation approaches make different assumptions about the underlying data [5]–[7].

1.2 Objective of the Project

Evaluate performance by analysing how imputation error scales with missingness rate for each mechanism (MCAR/MAR/MNAR) [5] and plot the reconstructed profiles against ground truth for each model. This gives the verdict about how each model works in different scenarios and the best performed models.

Section 2

2.1 Literature Review

Missing-data imputation has been studied extensively because incomplete observations occur in many real-world applications [5], [8]. The literature on missing-value recovery for electricity consumption data can be organised based on the methodology, which structures this project as a progression from *distribution-driven statistical approaches, through machine learning approaches, to representation-driven deep sequence models* [5], [6].

Statistical approaches may exploit distributional or temporal assumptions, machine-learning methods can learn nonlinear relationships, and deep-learning approaches can model more complex temporal features [2], [6].

An extended and broader review of missing-value processing emphasized the importance of missing mechanism, missing pattern and missing rate while imputing missing values [8].

2.2 Classical Statistical Foundational Methods implemented are:

- ***Multiple Iterative Chained Equations with Predictive Mean Matching*:** MICE PMM was implemented as a series of conditional regressions, repeated until the imputed distribution stabilised. Predictive mean matching kept the imputed values within realistic ranges [5], [8].
- ***Expectation Maximisation Algorithm*:** EM alternates between an E-step, which estimates the posterior over the missing values, and an M-step, which re-estimates the parameters by maximum

likelihood. Both Gaussian-mixture and log-normal variants were used to explore the change in performance with underlying distribution [8].

- ***Seasonal Kalman Smoothing*:** Load series is modelled with level, trend and daily seasonal components. A forward Kalman filter followed by a backward smoothing pass estimated the missing readings, so the daily cycle carried through the gaps [7].
- ***Low Rank Tensor Completion*:** The data were arranged as a meter × time × variable tensor. LRTC filled the gaps using low-rank structure across all three dimensions, and captured redundancy in time and correlations between electrical quantities [2], [3].

2.3 Machine Learning models implemented:

- ***MissForest*:** Treated imputation as a series of random-forest regressions, predicted each incomplete variable in turn without assuming a particular distribution [5].
- ***XGBoost*:** Each incomplete column was treated as a supervised target, with lag and rolling-window features engineered from the time series, which accounted for accuracy. [5].

2.4 Deep Learning models implemented are:

- ***Long Short Term Memory*:** The LSTM model processed daily sequences step by step. Its memory cell retained the typical load shape, and it fill gaps from temporal dependencies [6], [9].
- ***Bidirectional-Gated Recurrent Unit*:** Bi-GRU concatenated forward and backward hidden states, so each missing entry is estimated from both earlier and later readings, but gaps cause sequential discontinuities [2], [7], [9].
- ***Residual Network*:** 1-D ResNet built as window-based model with stacked convolutional blocks, skip connections. Trained to reconstruct masked entries directly, captured sharp load transitions.

Section 3

3.1 Overview of Methodology

The pipeline in Fig.1 shows methodology of the project to simulate realistic missingness in smart-meter readings dataset and then recover those missing values using statistical, machine-learning and deep learning imputation techniques, so that the accuracy of each technique can be judged against the known ground truth [5], [9].

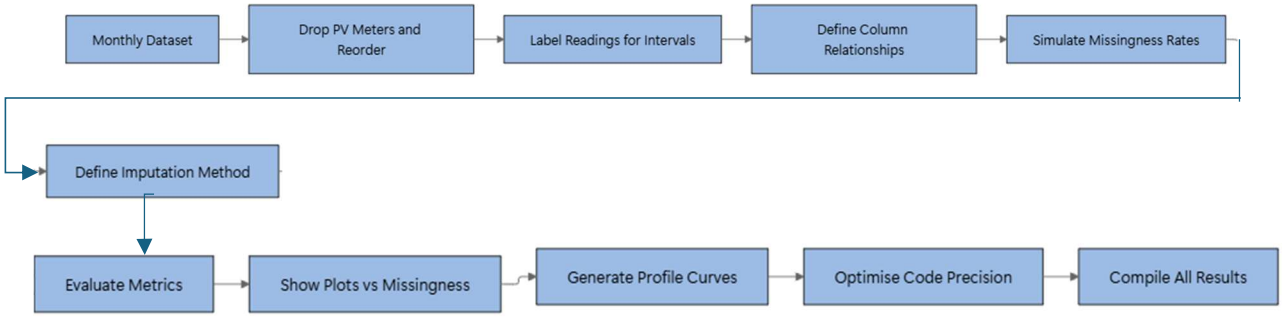


Fig. 1. Methodology pipeline

i) Missingness Mechanisms: Three missingness mechanisms are considered and simulated across all columns [5], [6], [8]:

- ***MCAR - Missing Completely At Random***, where missing values is independent of the observed and unobserved values.
- ***MAR - Missing At Random***, the probability of a value being missing is related to other observed variables.
- ***MNAR - Missing Not At Random***, the missingness mechanism is related to the value that is itself missing or information associated with it**.**

ii) Missingness Rates: For each missingness mechanism, multiple missingness percentages are generated. ***10%, 20%, 30%, 40%*** are the missingness rates considered for each missingness type and applied to the selected electrical variables [2], [5].

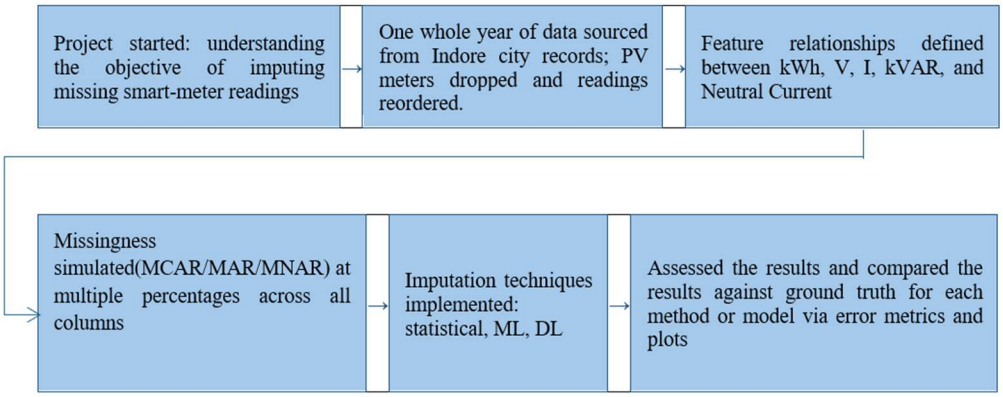
iii) Modelled electrical features: Multivariate relationships are captured between the electrical variables before applying the imputation algorithm and some constraints are applied to enforce real world electrical distribution laws [2].

- $S_{\text{proxy}} = \frac{V \times I \times 0.25}{1000}$ For estimating the apparent energy consumed over a 15-minute (0.25 hour) interval using instantaneous voltage and current measurements.
- $\Delta I = |I_{\text{phase}} - I_{\text{neutral}}|$ Captures potential leakage, load imbalance, or tamper conditions by measuring the absolute difference between the phase and neutral lines.
- $\theta = 2\pi \frac{t-1}{96}$ Converts the 96 daily 15-minute intervals (t) into continuous cyclical features to model the 24-hour load curve.
- $y_{\text{refined}} = \max(0, y_{\text{imputed}})$ Active Energy (kWh), Apparent Energy (kVAh), Phase Current (I_{phase}), and Neutral Current (I_{neutral}) are strictly bound to zero or positive values.
- $180 \leq V \leq 285$ Voltage (V) is capped to physical distribution limits to prevent anomalous extremes in the imputed data.
- $\text{If kWh} > \text{kVAh, then kWh} = \text{kVAh} \times 0.95$ Active power cannot exceed apparent power. If the algorithm outputs an invalid state where kWh > kVAh, the active energy is forcefully recalibrated to a fixed 0.95 power factor ratio.

iv) Implementing different Imputation Techniques: For each missingness scenario, the underlying mathematical formulation and working mechanism of the following imputation methods are studied and applied.

v) Evaluation and Visualisation: Error metrics are evaluated for each of the five quantities (kWh, V, I, kVAR, Neutral Current). *Metric vs missingness rate* plots are generated for each missingness mechanism (MCAR/MAR/MNAR), and *consumption vs 15-minute interval* profiles are generated for each mechanism and percentage, to visually trace how closely the imputed curve follows the ground-truth curve.

3.2 Work Executed Thus far:



Section 4

4.1 Comparison of all implemented models based on Metrics

In the below table, all the methods are ranked based on the metrics for imputation performance of missing values in kWh of MAR type with 30% missingness rate of a single month dataset.

	Model	Type	MAE	RMSE	MAPE%
1	***MissForest***	ML	**0.0040**	**0.0107**	24.16
2	***XGBoost***	ML	0.0041	0.0110	25.06
3	***EM-GMM***	Stat	0.0068	0.0128	**11.44**
4	***MICE-PMM***	Stat	0.0096	0.0187	16.12
5	***Seasonal Kalman***	Stat	0.0128	0.0309	21.38
6	***LRTC***	Stat	0.0253	0.0468	42.25
7	***GRU***	DL	0.0312	0.0650	30.33
8	***ResNet***	DL	0.0355	0.0691	34.68
9	***LSTM***	DL	0.0374	0.0727	36.53
10	***EM-Log Normal***	Stat	0.352	0.571	589.6

Mean Absolute Error (MAE): Measures the average magnitude of errors in original physical units [5], [7].

Root Mean Square Error (RMSE): Calculates the square root of average squared errors, heavily penalizing large deviations [4], [5]. Predicts unpredictable peak-load spikes.

Mean Absolute Percentage Error (MAPE): Expresses the average error as a relative percentage [2], it artificially inflates errors during near-zero load conditions.

Inferences obtained from the comparison table are:

MissForest: At 30% MAR missingness, had the lowest MAE (0.0040) and RMSE (0.0107). Captured nonlinear relationships well among the engineered features and followed load spikes. XGBoost was close behind (0.0041, 0.0110).

EM-GMM: Performed as the best after ML models, had the lowest MAPE (11.44%), even if its MAE (0.0068) was higher than the two tree-based methods.

GRU: The best deep-learning model (MAE 0.0312, RMSE 0.0650, MAPE 30.33%), but poorer than ML methods and EM-GMM. Because the masked gaps break the input sequences its recurrent layers depend on [2].

4.2 Daily Profile Plots of the best methods in each category

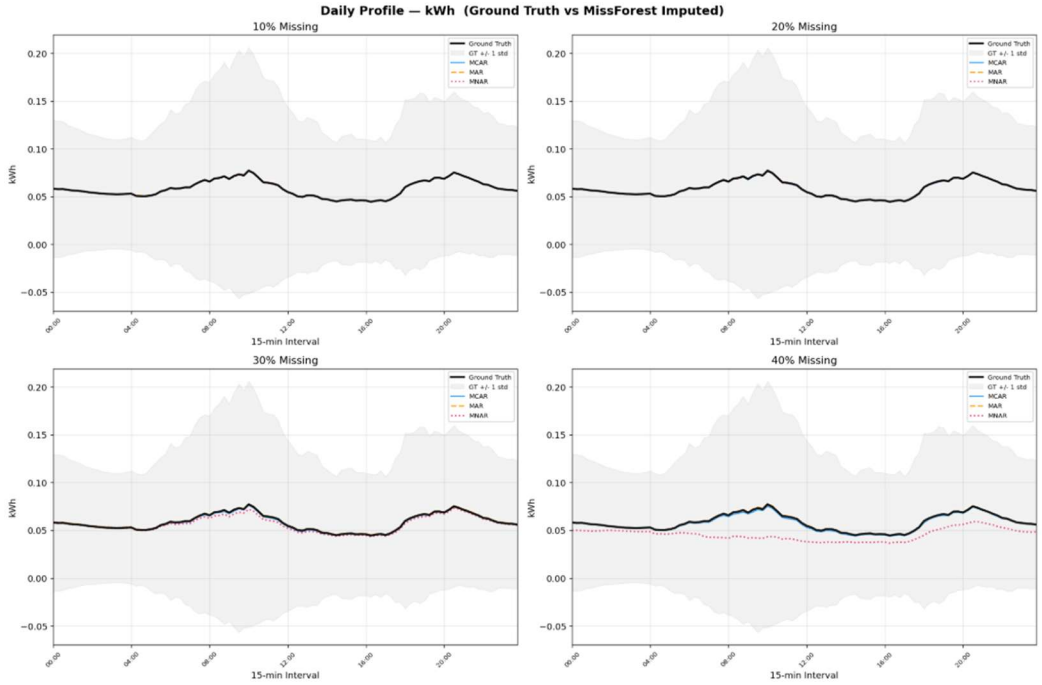


Fig. 2. Daily profile tracing of Nov month consumption kWh by MissForest model (ML)

At 10–30% missingness, the imputed curves for all three mechanisms nearly coincide with the ground truth. At 40%, MCAR and MAR stay close, while MNAR shows a slight underestimate of peak usage.

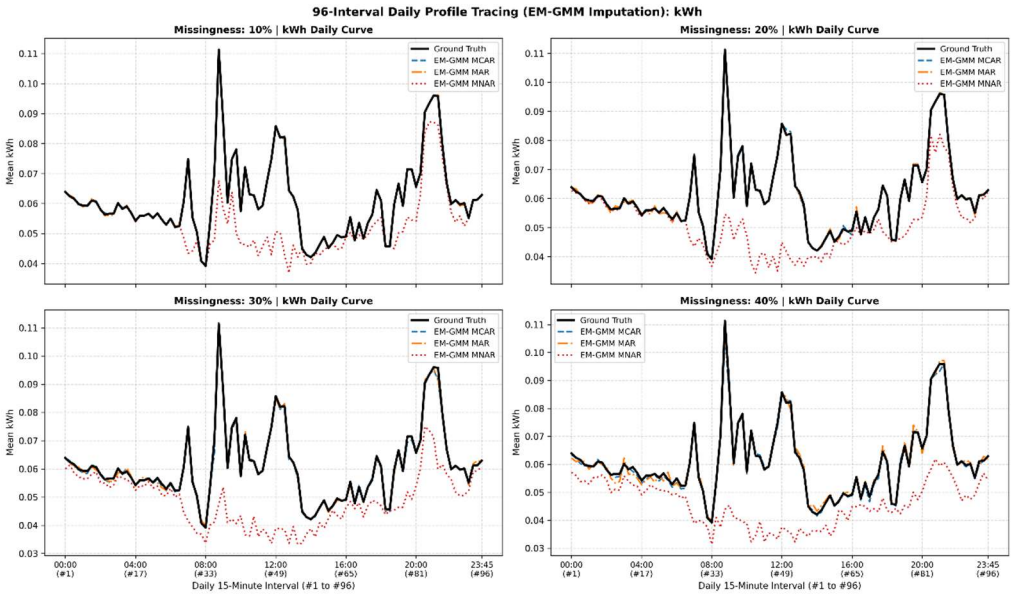


Fig. 3. Daily profile tracing of Oct month consumption kWh by EM GMM approach (Statistical)

Followed the ground truth closely under MCAR and MAR at every rate from 10% to 40%. Under MNAR it underestimated kWh at all rates, consistent with MNAR missingness depending on the missing value itself.

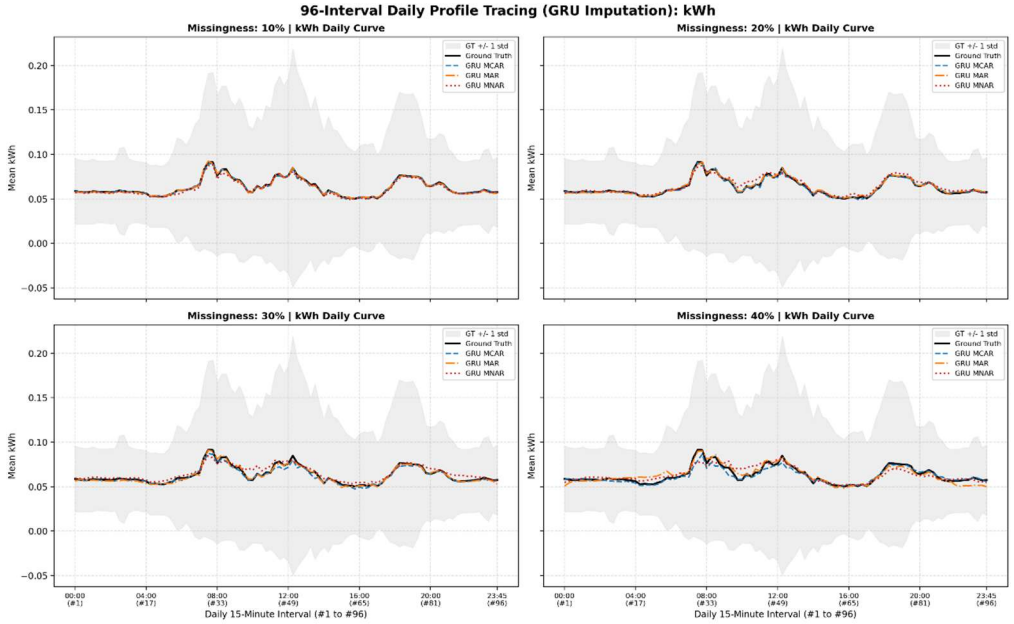


Fig. 4. Daily profile tracing of Nov month consumption kWh by GRU model (DL)

Didn't show the strong underestimation that EM-GMM showed under MNAR. However, due to gaps in the input sequences the curves were slightly smoothed at daytime peaks at 30% and 40% missingness.

4.3 Future Workflow Timeline

October - November

- Explore the performance of high-level breakthrough models in other fields, like GANs, Diffusion models in Smart meter imputation problem [1], [5], [10].
- Simulate higher missingness rates to see how the models cope when large blocks of data are missing [1], [5].
- Run imputation on all 12 months in one pass and tune the models accordingly.
- Keep refining the code so the imputed curve follows the ground truth more closely.
- Compile the comparative inferences and results across all techniques into the final report

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